Hearing Drones in 3D: Convolutional Detection and Direction-Finding from a Microphone Array

Shai Livshits, Dan Masad

Project Category: Applied – Audio / Convolutional Networks / Defense & Security

Problem Definition

Drones are hard to see and often fly with no radio link, defeating cameras and radio detectors. Sound is a cheap fallback: rotors are loud and distinctive. From a ground microphone array we want a network reporting ten times per second whether a drone is audible and where: azimuth (compass bearing) and elevation (angle above the horizon).

Key Challenges

A distant drone is quiet and easily confused with wind, traffic, and aircraft. It fills a tiny fraction of the search space, so "nothing" scores well – severe class imbalance. Recordings are continuous: neighbouring windows look almost identical, so a random split would leak training data into test.

Data

Western Sydney University's *UAV Acoustic Localization Dataset* (CC BY-SA 4.0) downloads freely: ~75 minutes of a DJI Air 3 at two sites, ~10 minutes of drone-free background, with GPS logs every 100 ms giving direction and speed: labels come free. The 24 microphones sit on a 1.3 m tripod wired as four synchronised six-channel recorders. A downloaded slice confirmed labels and geometry agree; we train on one site, test on another.

Proposed Method

From sound to images: Sound from one direction reaches the microphones at slightly different times. Delaying each by those offsets and summing reinforces a correct guess and partly cancels a wrong one. Sweeping the dataset's direction grid turns each 100 ms of audio into a 91×23 image of energy per direction; the drone is a bright blob. Fixed signal processing turns this into an image problem.

Models: The baseline picks the brightest pixel, as classical systems do, failing on noise and at range. We train, from scratch, an encoder–decoder CNN outputting a per-pixel drone probability plus one "drone present" score, with class-weighted cross-entropy and Adam. Consecutive images enter as extra input channels. A secondary output regresses speed from Doppler shift: an approaching drone's tone rises.

Evaluation

On the unseen site: precision–recall and false alarms per hour for detection; median angular error and hit rates at the 10° the dataset itself calls correct; mean absolute error for speed. We repeat everything per subset – one recorder at a time, then six picked for maximum spread – and sweep frame-stack length.

Related Work and Contribution

Rodríguez et al. (2026) released this dataset and a single-frame model, naming temporal context as future work and publishing no code; Al-Emadi et al. (2019) detect from one microphone, without direction. Our question is practical: how little array does this task need? Six of the 24 microphones already span the whole 1.3 m, so the other 18 buy no angular resolution – only suppression of the false peaks a sparse array invents. We hypothesise that temporal context supplies it instead: a real drone drifts smoothly while false peaks flicker – making cheap, widely spaced arrays operationally practical. Two side questions follow: whether their reported drop in false alarms over classical beamforming survives a proper test (theirs used five minutes of drone-free audio) and whether Doppler yields speed.

References

1. B. J. Rodríguez, S. F. Chevtchenko, M. Herrera Martínez, Y. Bethi, S. Afshar. *Acoustic Imaging for UAV Detection: Dense Beamformed Energy Maps and U-Net SELD.* arXiv:2508.00307, Aug 2025 (rev. Jul 2026); IWAENC 2026.
2. B. J. Rodríguez, S. F. Chevtchenko, S. Afshar. *UAV Acoustic Localization Dataset: 24-Channel Beamformed Recordings of a DJI Air 3 Drone with Synchronized GPS Flight Logs.* Western Sydney University, published 15 Jul 2026. DOI 10.26183/07nm-4p35.
3. S. Al-Emadi, A. Al-Ali, A. Mohammad, A. Al-Ali. *Audio Based Drone Detection and Identification using Deep Learning.* IWCMC, 2019.